



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°1_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

PRÉNOM: Lens Sandro

NOM: PÉTIOTE

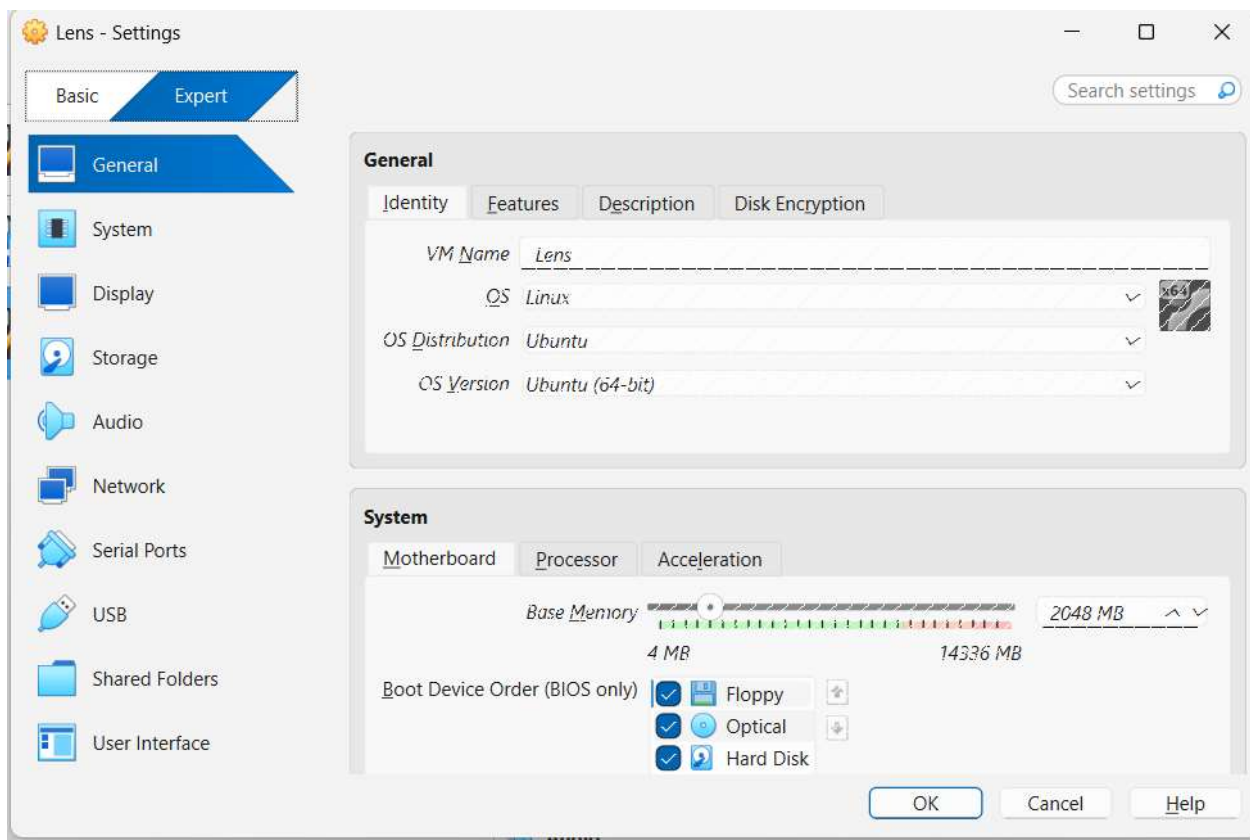
NIVEAU: L3

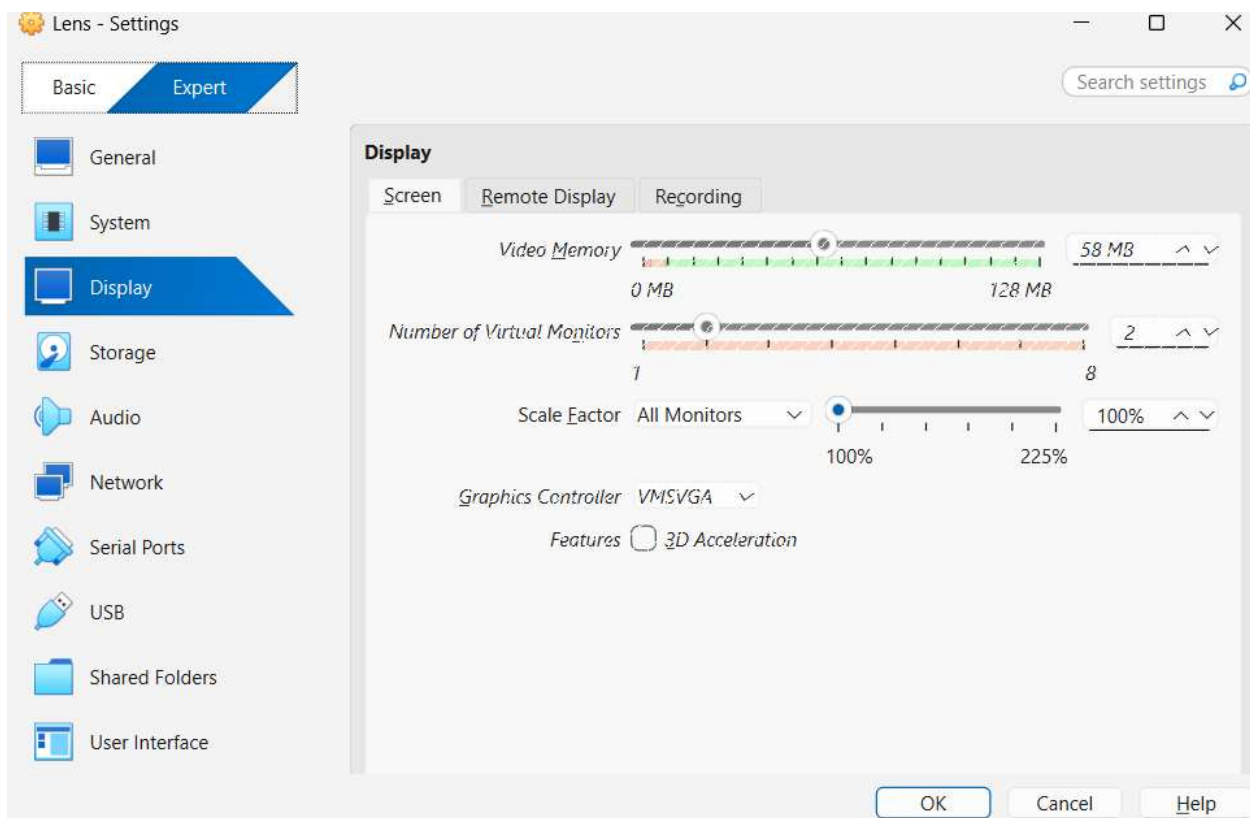
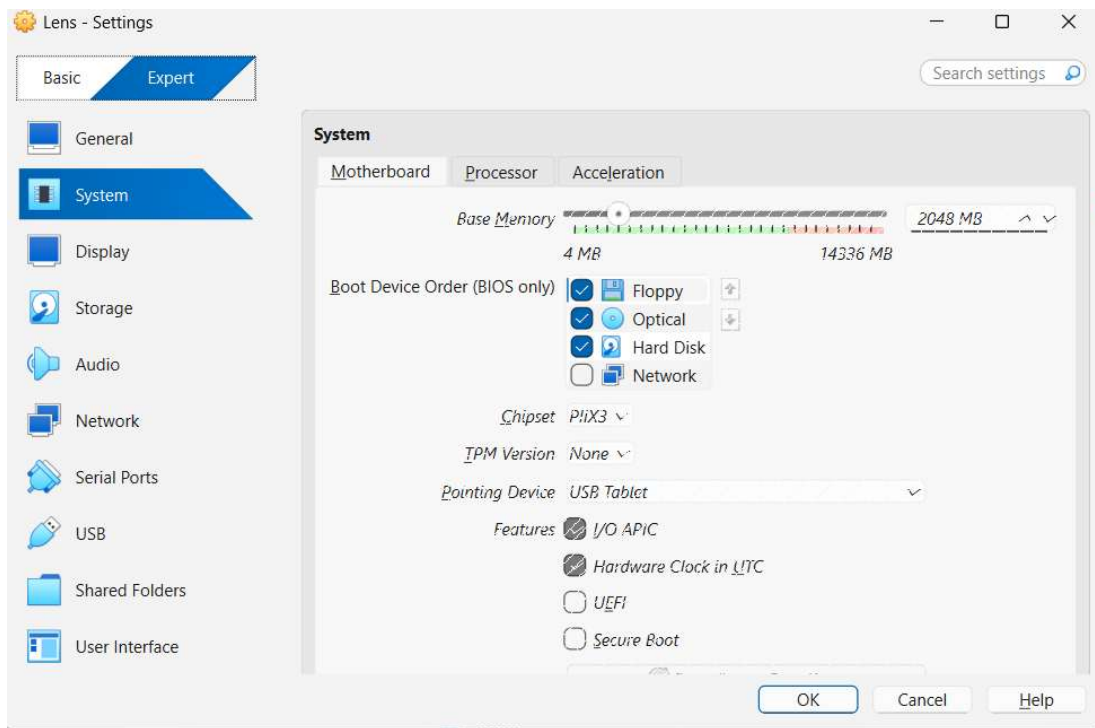
DATE: 07/03/2026

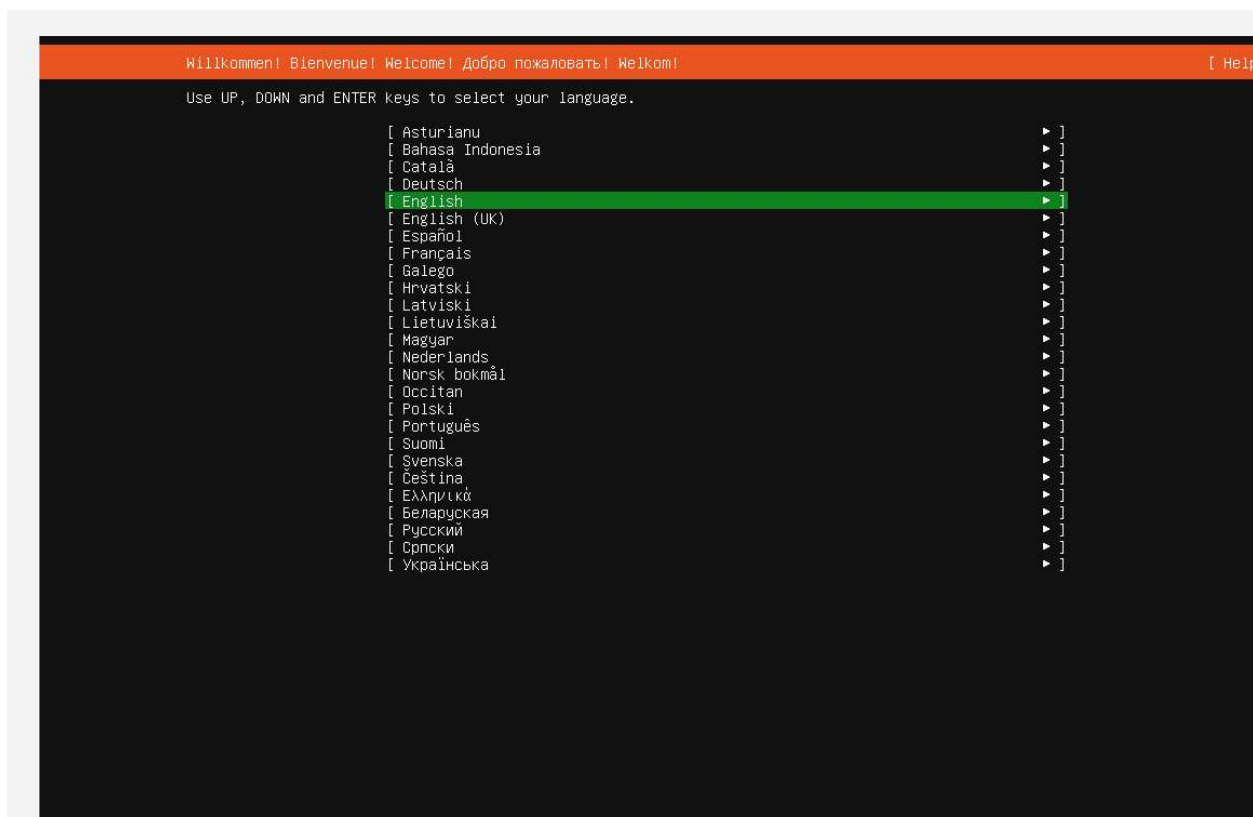
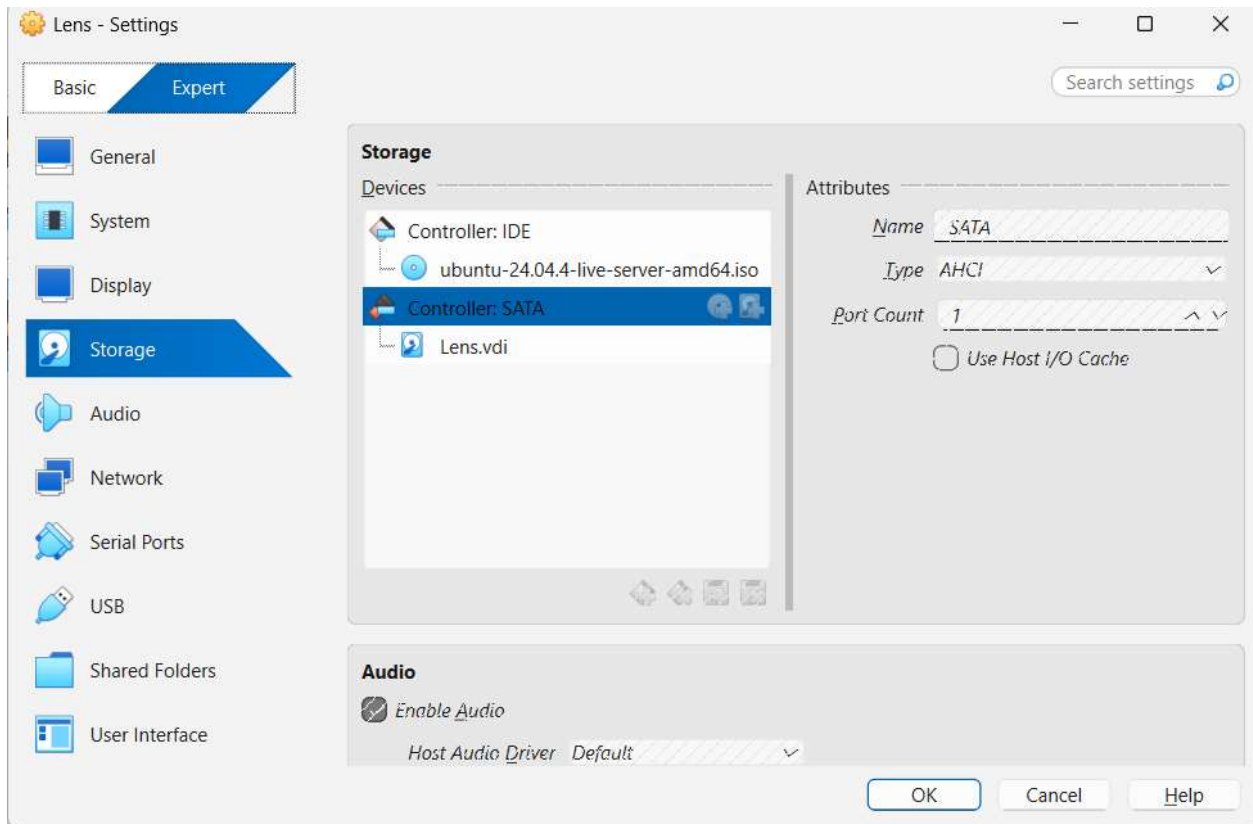
L'objectif de ce TD est de:

1. Installer un serveur Linux minimal
2. Configurer le réseau
3. Mettre en place l'accès distant sécurisé avec SSH
4. Effectuer la configuration de base d'un serveur

1- Installation de Ubuntu serveur et creation de la machine







Keyboard configuration

[Help]

Please select your keyboard layout below, or select "Identify keyboard" to detect your layout automatically.

Layout: [English (US) ▼]

Variant: [English (US) ▼]

[Identify keyboard]

[Done]

[Back]

Choose the type of installation

[Help]

Choose the base for the installation.

(X) Ubuntu Server

The default install contains a curated set of packages that provide a comfortable experience for operating your server.

() Ubuntu Server (minimized)

This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.

Additional options

[] Search for third-party drivers

This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.

[Done]

[Back]

Network configuration

[Help]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.

NAME	TYPE	NOTES
[enp0e3 eth ~ ▶]		
DHCPv4	10.0.2.15/24	08:00:27:17:d8:31 / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)
[Create bond ▶]		

[Done]

[Back]

Proxy configuration

[Help]

If this system requires a proxy to connect to the internet, enter its details here.

Proxy address:

If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.

The proxy information should be given in the standard form of "http://[user][:pass]@host[:port]/".

[Done]

[Back]

Ubuntu archive mirror configuration

[Help]

If you use an alternative mirror for Ubuntu, enter its details here.

Mirror address:

You may provide an archive mirror to be used instead of the default.

The mirror location is being tested. |

Done

Back

Guided storage configuration

[Help]

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[VBOX_HARDDISK_VB56b58652-17f99c8e local disk 25.000G ▼]

[X] Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

[] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

() Custom storage layout

Done

Back

Storage configuration [Help]

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
[/	11.496G	new ext4	new LVM logical volume ▶]
[/boot	2.000G	new ext4	new partition of local disk ▶]

AVAILABLE DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	22.996G ▶]
free space		11.500G ▶]

[Create software RAID (md) ▶]

[Create volume group (LVM) ▶]

USED DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	22.996G ▶]
ubuntu-lv	new, to be formatted as ext4, mounted at /	11.496G ▶]
[VBOX_HARDDISK_VBS6b58652-17f99c8e	local disk	25.000G ▶]
partition 1	new, BIOS grub spacer	1.000M ▶]
partition 2	new, to be formatted as ext4, mounted at /boot	2.000G ▶]
partition 3	new, PV of LVM volume group ubuntu-vg	22.997G ▶]

[Done]

[Reset]

[Back]

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Sandro

Your servers name: lens

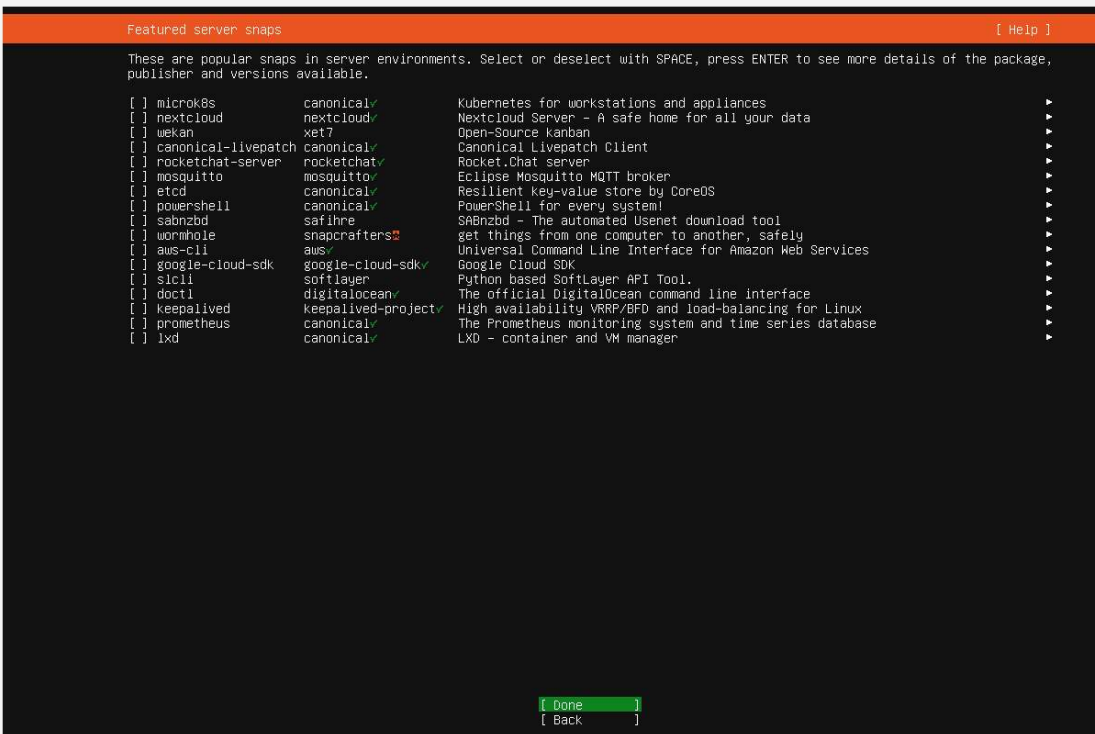
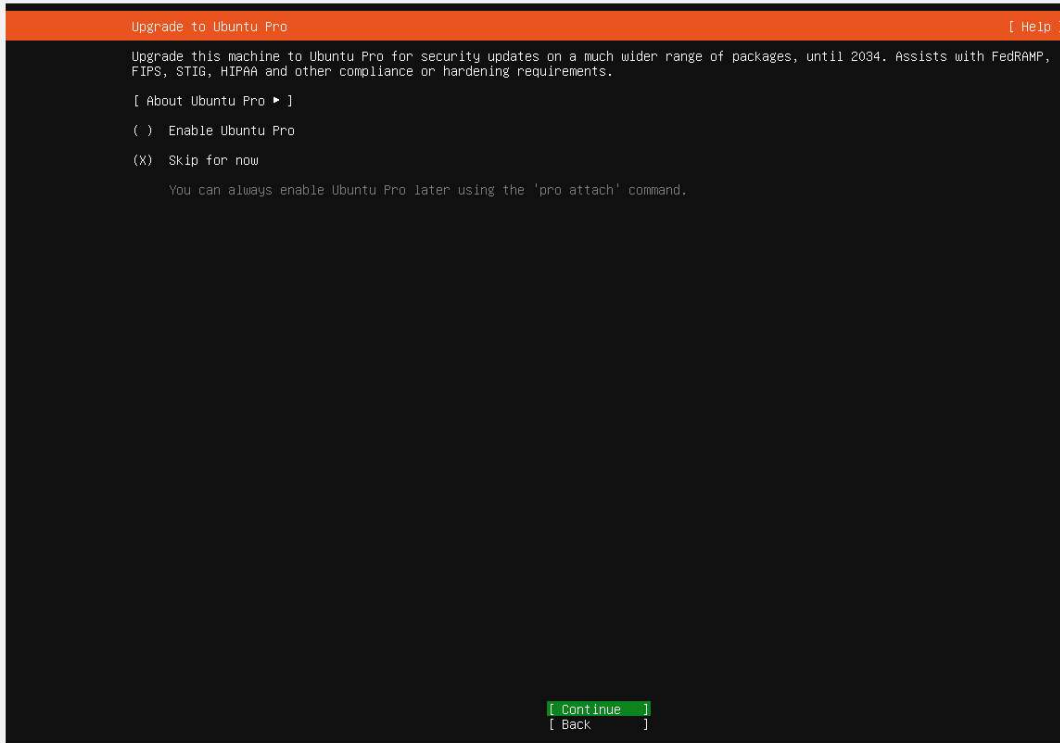
The name it uses when it talks to other computers.

Pick a username: sandro

Choose a password: ****

Confirm your password: ****

[Done]



2- Depuis votre machine physique, exécutez ces commandes via PowerShell.

- Connection à mon serveur depuis ma machine physique et test de quelques commandes.

```
PS C:\Users\Lens Sandro Petiot> ssh lens@172.20.10.4
The authenticity of host '172.20.10.4 (172.20.10.4)' can't be established.
ED25519 key fingerprint is SHA256:nODC/5aoIY4E9BhwtIzMjRr6IoPB7ncus2ADLZNPqBw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.20.10.4' (ED25519) to the list of known hosts.
lens@172.20.10.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Mar  8 06:02:44 PM UTC 2026

System load:  0.31               Processes:            103
Usage of /:   40.7% of 11.21GB   Users logged in:     1
Memory usage: 9%                IPv4 address for enp0s3: 172.20.10.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

lens@lens:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 24.04.4 LTS
Release:       24.04
Codename:      noble
lens@lens:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:17:d8:31 brd ff:ff:ff:ff:ff:ff
    inet 172.20.10.4/28 metric 100 brd 172.20.10.15 scope global dynamic enp0s3
        valid_lft 86116sec preferred_lft 86116sec
    inet6 fe80::a00:27ff:fe17:d831/64 scope link
        valid_lft forever preferred_lft forever
lens@lens:~$ ping -c 4 google.com
PING google.com (173.194.215.101) 56(84) bytes of data:
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=1 ttl=104 time=92.8 ms
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=2 ttl=104 time=97.1 ms
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=3 ttl=104 time=92.1 ms
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=4 ttl=104 time=87.5 ms
```

64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=4 ttl=104 time=87.5 ms

--- google.com ping statistics ---

4 packets transmitted, 4 received, 0% packet loss, time 3003ms

rtt min/avg/max/mdev = 87.541/92.383/97.067/3.377 ms

lens@lens:~\$ df -h

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	197M	1.1M	196M	1%	/run
/dev/mapper/ubuntu--vg-ubuntu--lv	12G	4.6G	6.1G	43%	/
tmpfs	985M	0	985M	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/sda2	2.0G	104M	1.7G	6%	/boot
tmpfs	197M	12K	197M	1%	/run/user/1000

lens@lens:~\$ free -h

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	318Mi	1.5Gi	1.0Mi	208Mi	1.6Gi
Swap:	2.0Gi	0B	2.0Gi			

lens@lens:~\$ sudo systemctl status ssh

[sudo] password for lens:

● ssh.service - OpenBSD Secure Shell server

Loaded: loaded (/usr/lib/systemd/system/ssh.service; **enabled**; preset: **enabled**)

Active: **active (running)** since Sun 2026-03-08 17:59:56 UTC; 6min ago

TriggeredBy: ● ssh.socket

Docs: man:sshd(8)

man:sshd_config(5)

Process: 700 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)

Main PID: 712 (sshd)

Tasks: 1 (limit: 2263)

Memory: 4.1M (peak: 5.1M)

CPU: 341ms

CGroup: /system.slice/ssh.service

└─712 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 08 17:59:53 lens systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...

Mar 08 17:59:56 lens sshd[712]: Server listening on 0.0.0.0 port 22.

Mar 08 17:59:56 lens sshd[712]: Server listening on :: port 22.

Mar 08 17:59:56 lens systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

Mar 08 18:02:44 lens sshd[940]: Accepted password for lens from 172.20.10.2 port 64872 ssh2

Mar 08 18:02:44 lens sshd[940]: pam_unix(sshd:session): session opened for user lens(uid=1000) by lens(uid=0)

lens@lens:~\$ ps

PID	TTY	TIME	CMD
996	pts/0	00:00:00	bash
1036	pts/0	00:00:00	ps

lens@lens:~\$ top

top - 18:06:57 up 6 min, 2 users, load average: 0.08, 0.20, 0.14
Tasks: 97 total, 1 running, 96 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 6.2 sy, 0.0 ni, 93.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1968.1 total, 1586.3 free, 317.1 used, 210.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 1651.0 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
995	lens	20	0	15128	7116	5148	S	4.5	0.4	0:02.89	sshd
1	root	20	0	22116	13248	9476	S	0.0	0.7	0:05.87	systemd

top - 18:06:58 up 6 min, 2 users, load average: 0.08, 0.20, 0.14
Tasks: 97 total, 2 running, 95 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 2.5 sy, 0.0 ni, 95.0 id, 0.0 wa, 0.0 hi, 2.5 si, 0.0 st

- Visualisation de mon systeme en temps réel avec TOP

```
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```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
55	root	20	0	0	0	0	I	1.3	0.0	0:04.70	kworker/0:2-events
995	lens	20	0	15128	7116	5148	S	0.3	0.4	0:03.03	sshd

```
top - 18:08:07 up 8 min, 2 users, load average: 0.02, 0.16, 0.13
Tasks: 97 total, 1 running, 96 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.9 sy, 0.0 ni, 98.7 id, 0.0 wa, 0.0 hi, 0.4 si, 0.0 st
MiB Mem : 1968.1 total, 1586.3 free, 317.1 used, 210.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 1651.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
55	root	20	0	0	0	0	I	1.0	0.0	0:05.49	kworker/0:2-events
649	message+	20	0	9780	5484	4660	S	0.3	0.3	0:00.20	dbus-daemon
1	root	20	0	22116	13248	9476	S	0.0	0.7	0:05.87	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.56	kworker/0:1-cgroup_destroy
11	root	20	0	0	0	0	I	0.0	0.0	0:02.36	kworker/u2:0-events_power_effici
12	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	20	0	0	0	0	S	0.0	0.0	0:00.37	ksoftirqd/0
17	root	20	0	0	0	0	I	0.0	0.0	0:00.20	rcu_preempt
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
22	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd

Coclusion.

Ce TD m'a permis de savoir comment installer un serveur Linux minimal, configure le réseau, mettre en place l'accès distant sécurisé avec SSH et effectuer la configuration de base d'un serveur, donc la tache a été réussie.